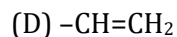
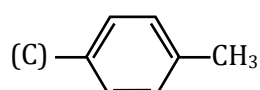
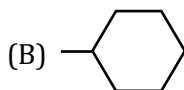
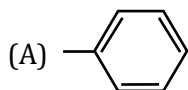




Electronic Displacement Effect : RACE-01

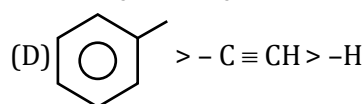
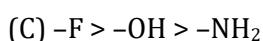
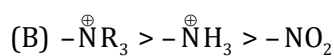
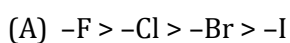
**Inductive effect (Groups showing +I and -I effects)**

1. Which of the following groups have + I effect :



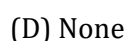
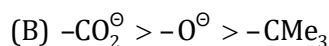
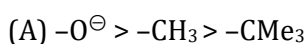
CED089

2. Which of the following is false order of - I effect ?



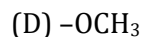
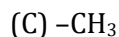
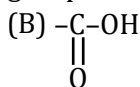
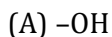
CED102

3. What is the correct order of + I effect ?



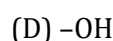
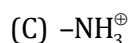
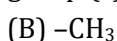
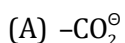
CED103

4. Which of the following groups have -I effect :



CED107

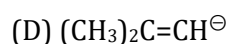
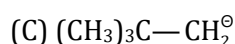
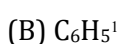
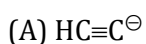
5. Which of the following group (s) exerts +I effect ?



CED245

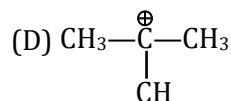
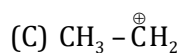
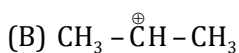
**Stability of Intermediates**

6. Most stable carbanion is :-



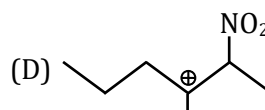
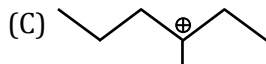
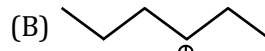
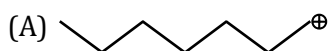
CED090

7. Which of the following is most stable carbocation ?



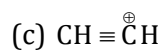
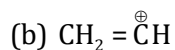
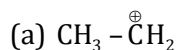
CED104

8. The most stable carbocation among the following :



CED105

9. Give stability in increasing order of following intermediates:



(A)  $a > b > c$

(B)  $c > b > a$

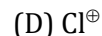
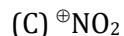
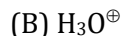
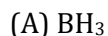
(C)  $c > a > b$

(D)  $b > a > c$

CED106

**Electrophiles and Nucleophiles**

10. Which of the following species is not electrophilic in nature :-



CED096

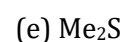
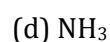
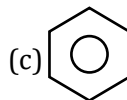
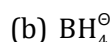
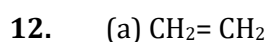
11. Wrong statement regarding methyl carbonium ion  $\overset{\oplus}{\text{C}}\text{H}_3$ .

(A) It is  $\text{sp}^2$  hybridised(B) Vacant orbital is  $\text{sp}^2$  hybridised

(C) Vacant orbital is perpendicular to molecular planar and in pure p-orbital

(D) It is electrophile with sextet of electron

CED101



Above are examples of some Nucleophiles if

X = Number of compounds in which  $\pi$  bonding orbital is HOMO.

Y = Number of compounds in which  $\sigma$  bonding orbital is HOMO.

Z = Number of compounds in which Non-bonding (L.P) is HOMO.

Then  $X + Y - Z = ?$

CED364

**ANSWER KEY**

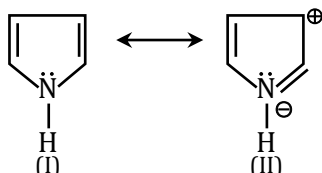
| Question | 1  | 2  | 3 | 4   | 5  | 6 | 7 | 8 | 9 | 10 |
|----------|----|----|---|-----|----|---|---|---|---|----|
| Answer   | B  | D  | C | ABD | AB | A | D | C | A | B  |
| Question | 11 | 12 |   |     |    |   |   |   |   |    |
| Answer   | B  | 1  |   |     |    |   |   |   |   |    |



Electronic Displacement Effect : RACE-02

*Resonating structures and their Stabilities*

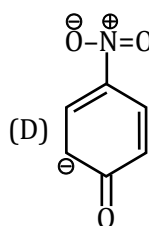
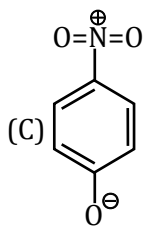
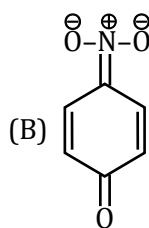
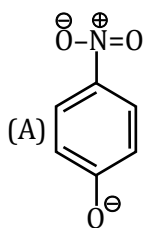
1. Examine the following two structures for pyrrole and choose the correct statement given below



- (A) II is not an acceptable resonating structure because carbonium ions is less stable than nitride ion  
 (B) II is not an acceptable resonating structure because there is charge separation  
 (C) II is not an acceptable resonating structure because nitrogen has ten valence electrons  
 (D) II is an acceptable resonating structure

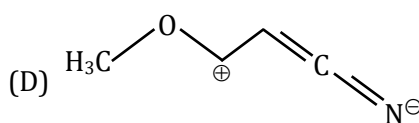
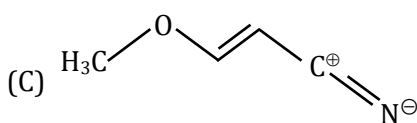
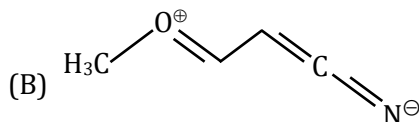
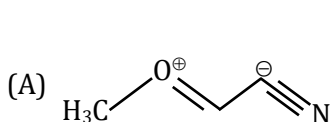
CED118

2. The most unlikely representation of resonance structure of p-nitrophenoxide ion is-



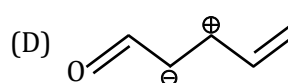
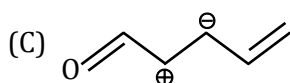
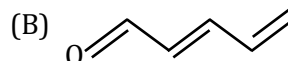
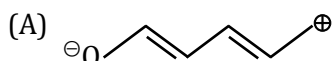
CED120

3. Which of the following structure is the most stable ?



CED125

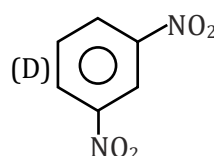
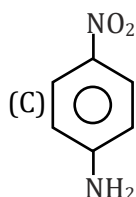
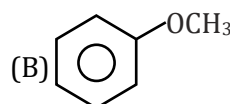
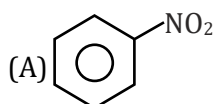
4. The least stable resonating structure is



CED126

**Mesomeric effect (Groups showing +M and -M effects)**

5. In which of the following molecules  $\pi$  - electron density in ring is minimum:

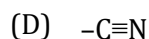
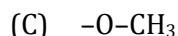
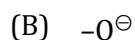
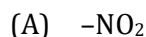


CED133

6. Match the column I with column II.

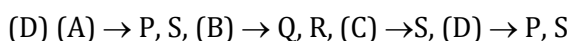
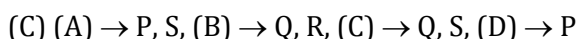
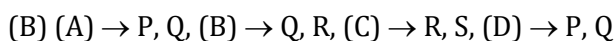
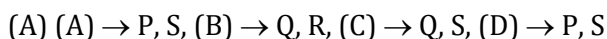
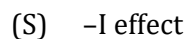
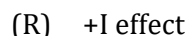
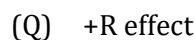
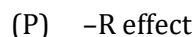
**Column-I**

(Group attached with benzene ring)



**Column-II**

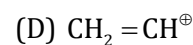
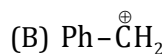
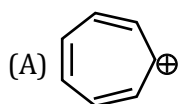
(Effect shown by the group)



CED138

**Aromaticity**

7. Among the following the most stable cation is



CED150

8. STATEMENT-I : Tropylium cation is more stable than  $(\text{CH}_3)_3\text{C}^\oplus$

STATEMENT-II : Tropylium cation has aromatic character

(A) Statement-I is true, Statement-II is true; Statement-II is a correct explanation for statement-I

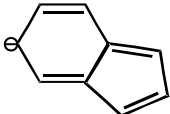
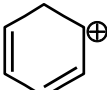
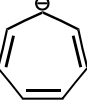
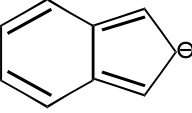
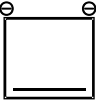
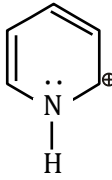
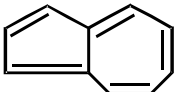
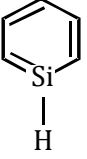
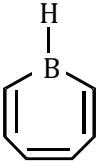
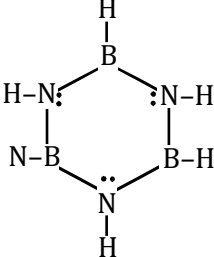
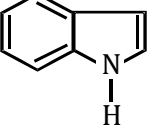
(B) Statement-I is true, Statement-II is true; Statement-II is not a correct explanation for statement-I

(C) Statement-I is true, Statement-II is false

(D) Statement-I is false, Statement-II is true

CED151

9. Select the set containing all aromatic species.

- (A) (a)  (b)  (c) 
- (B) (a)  (b)  (c)  (d) 
- (C) (a)  (b)  (c)  (d) 
- (D) (a)  (b)  (c)  (d) 
- (e)  (f) 

CED165

10. Non-aromatic compound is/are ?

- (A)  (B)  (C)  (D) All of these

CED166

ANSWER KEY

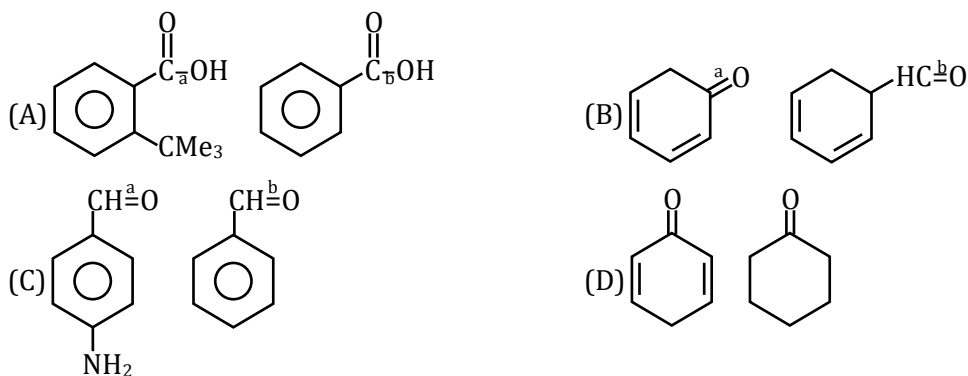
| Question | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|----------|---|---|---|---|---|---|---|---|---|----|
| Answer   | C | C | B | C | D | A | A | A | D | AB |



## Electronic Displacement Effect : RACE-03

**Bond length and Bond Strength**

1. In which case, C - O bond length is shorter for 1<sup>st</sup> compound :



CED175

2. Which of the following is incorrect order of bond length



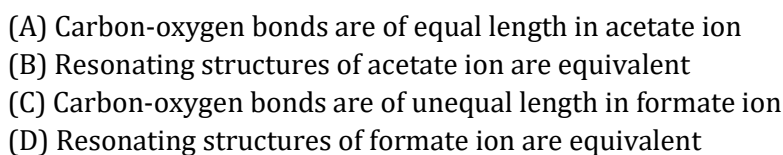
CED183

3. Which of the following C-H bonds shown would have the smallest bond dissociation energy?



CED240

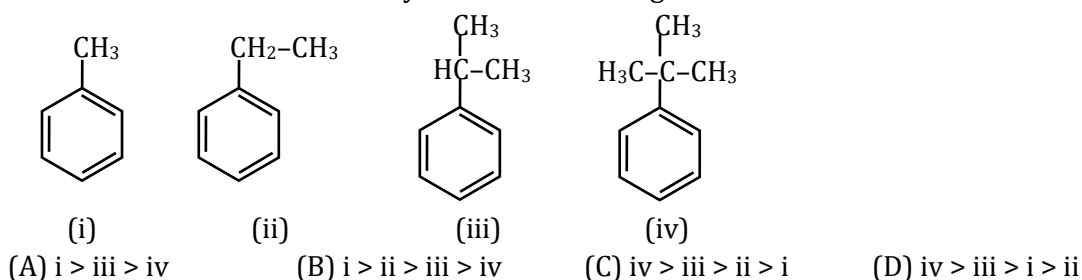
4. Select correct statement :



CED188

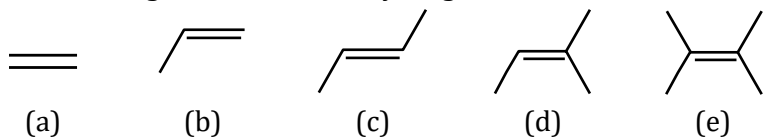
**Applications of Hyperconjugation**

5. Correct order of electron density in the benzene ring is :



CED194

6. Give increasing order of heat of hydrogenation:



(A)  $a > c > d > e > b$

(B)  $a > d > c > e > b$

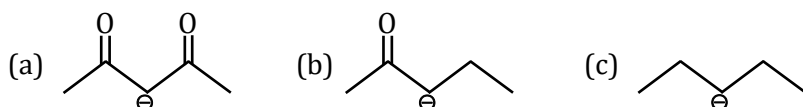
(C)  $a > b > c > d > e$

(D)  $b > a > c > d > e$

CED205

### Stability of Intermediates

7. Correct order of intermediates in increasing order of their stability.



(A)  $a > b > c$

(B)  $c > b > a$

(C)  $b > a > c$

(D)  $a > c > b$

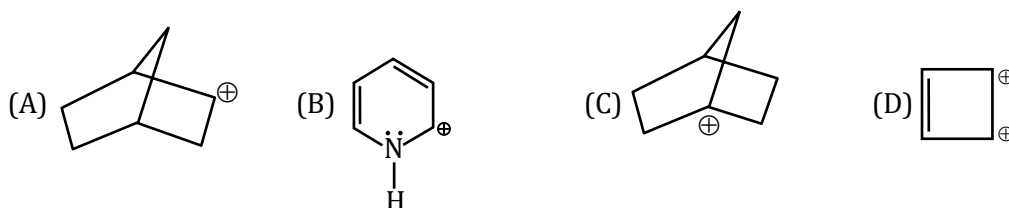
CED197

8. Among given cations which is most stable



CED198

9. Among the following, the least stable cation is :



CED200

### ANSWER KEY

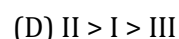
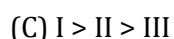
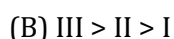
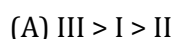
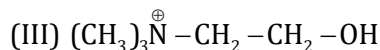
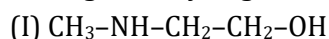
| Question | 1 | 2 | 3 | 4   | 5 | 6 | 7 | 8 | 9 |
|----------|---|---|---|-----|---|---|---|---|---|
| Answer   | A | C | B | ABD | B | C | A | A | C |



## Electronic Displacement Effect : RACE-04

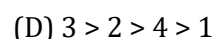
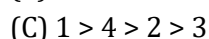
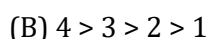
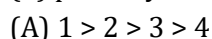
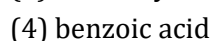
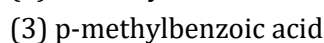
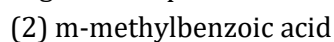
## Acidic strength

1. Arrange acidity of given compounds in decreasing order:



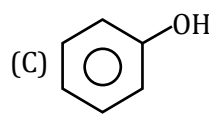
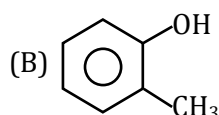
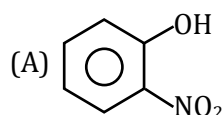
CED213

2. Select correct order regarding acidic strength of given compounds :



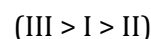
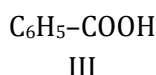
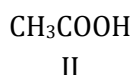
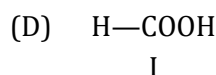
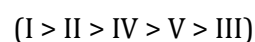
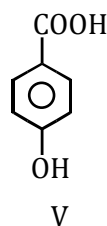
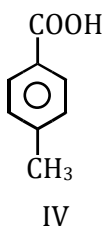
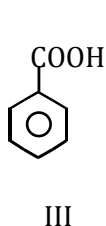
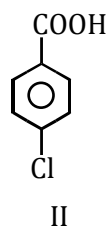
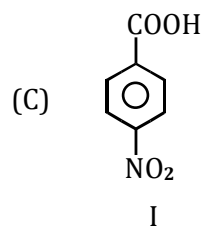
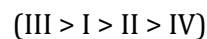
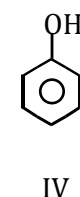
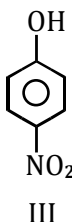
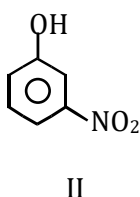
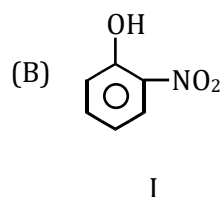
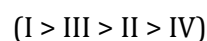
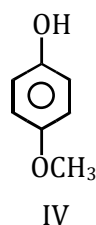
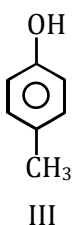
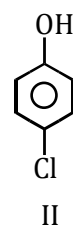
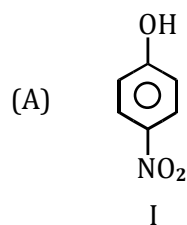
CED216

3. Which one of the following compounds is most acidic:-



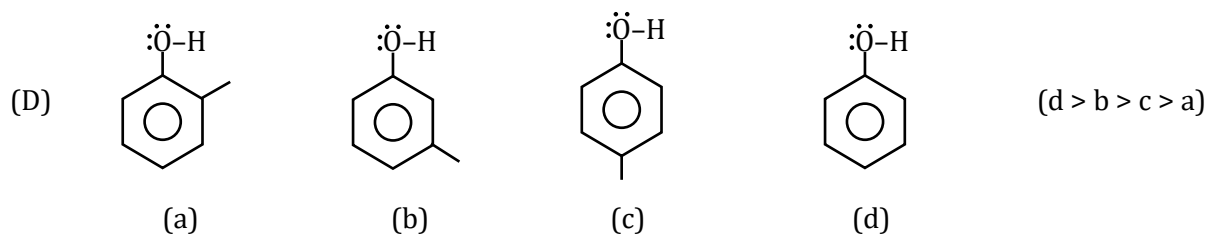
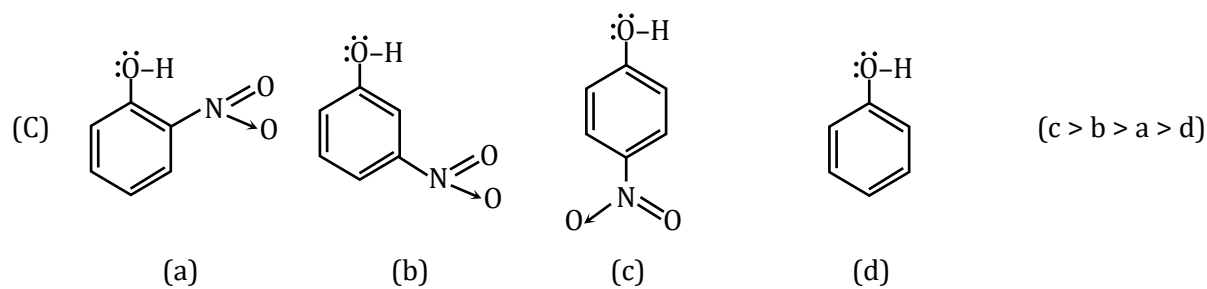
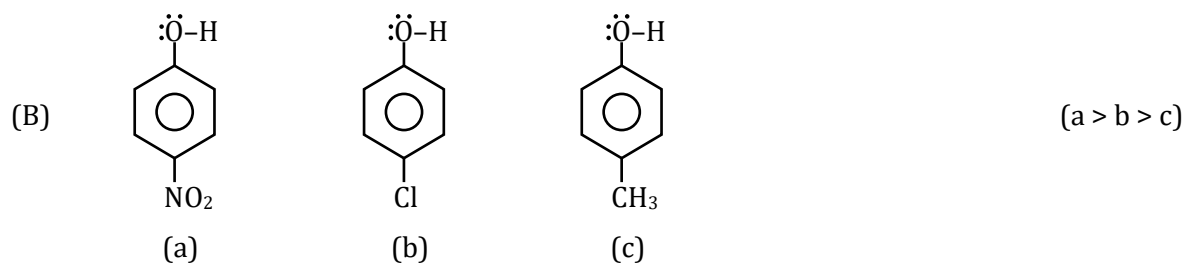
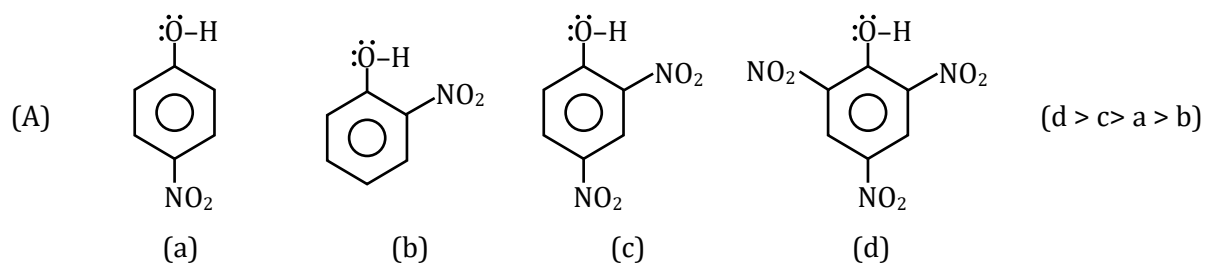
CED220

4. Correct set of acidic strength order in given option is :



CED221

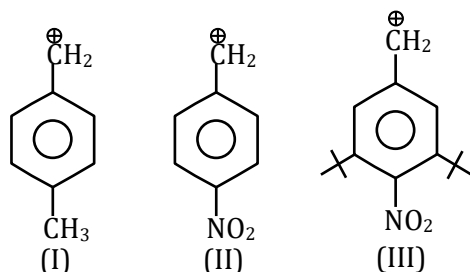
5. Incorrect set of acidic strength order in given option is :



CED222

### SIR Effect

6. Arrange the following carbocation in the increasing order of stability :



(A) I < II < III

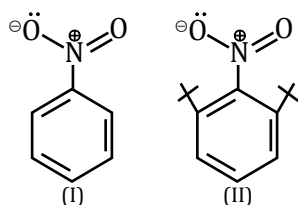
(B) II < III < I

(C) III < II < I

(D) III < I < II

CED226

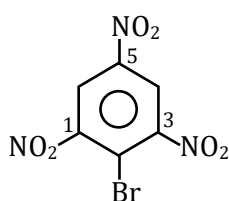
7. Consider the following two structures and choose the correct statements –



- (A) carbon-nitrogen bond length structure I is greater than that in structure II  
 (B) carbon-nitrogen bond length in structure I is less than in structure II  
 (C) carbon-nitrogen bond length in both structure is same  
 (D) It can not be compared

CED176

8. Which of the following statements would be true about this compound :

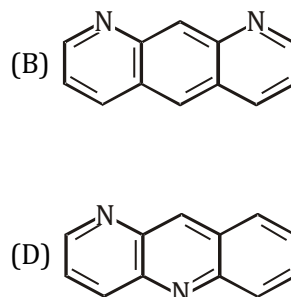
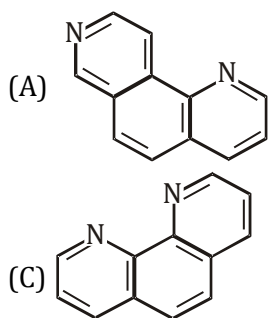


- (A) All three C-N bonds are of same length.  
 (B) C<sub>1</sub>-N and C<sub>3</sub>-N bonds are of same length but shorter than C<sub>5</sub>-N bond.  
 (C) C<sub>1</sub>-N and C<sub>3</sub>-N bonds are of same length but longer than C<sub>5</sub>-N bond.  
 (D) C<sub>1</sub>-N and C<sub>3</sub>-N bonds are of different length but both are longer than C<sub>5</sub>-N bond

CED177

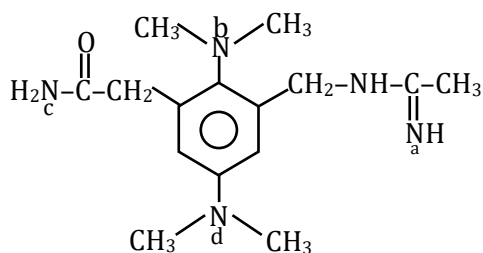
### Basic strength

9. Which of the following is most basic:



CED234

10. Basicity order of N in following compound is :



- (A)  $b > d > a > c$       (B)  $a > b > d > c$       (C)  $a > b > c > d$       (D)  $a > c > b > d$

CED235

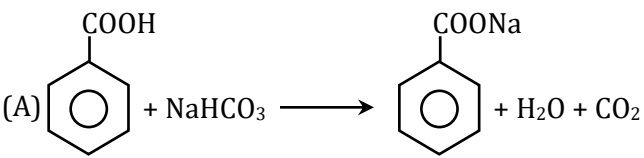
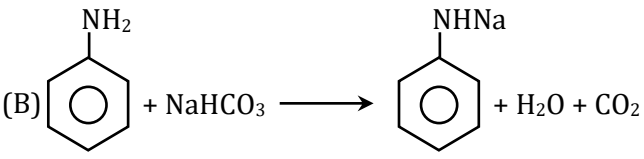
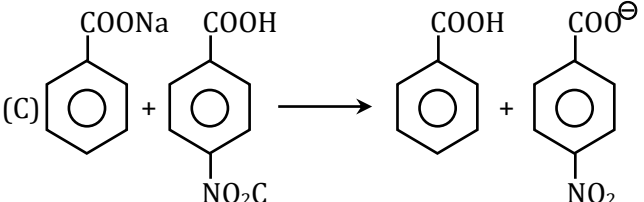
11. Which of the following is the weakest base?

- (A)  $\text{Cl}^\ominus$  (B)  $\text{F}^\ominus$  (C)  $\text{OH}^\ominus$  (D)  $\text{CH}_3^\ominus$

CED112

**Acid Base Equilibrium**

12. Identify the feasible reactions

- (A)    
 (B)    
 (C)    
 (D) Both A and C

CED239

**ANSWER KEY**

| Question | 1  | 2  | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|----------|----|----|---|---|---|---|---|---|---|----|
| Answer   | A  | C  | A | B | C | B | B | C | C | B  |
| Question | 11 | 12 |   |   |   |   |   |   |   |    |
| Answer   | A  | D  |   |   |   |   |   |   |   |    |



## Electronic Displacement Effect : RACE-05

## Quantitative Analysis

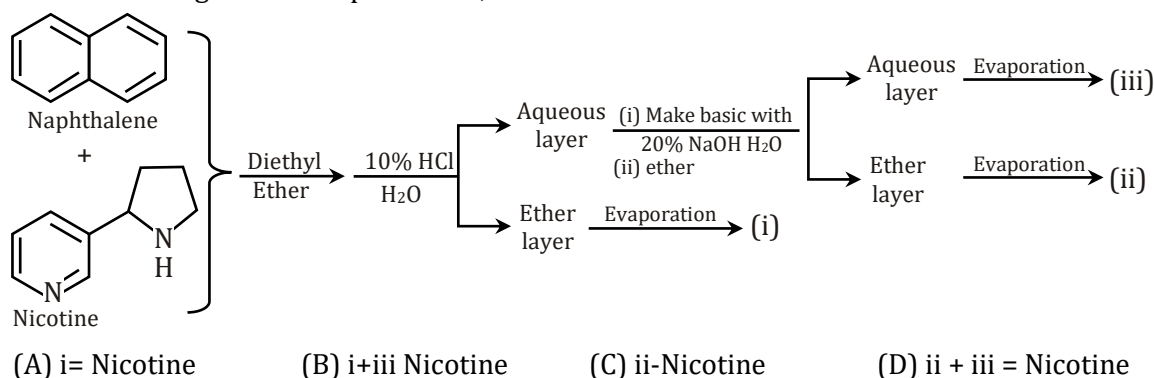
- In Kjeldahl's method, nitrogen present is estimated as :  
(A)  $N_2$  (B)  $NH_3$  (C)  $NO_2$  (D) none of these  
**CED365**
- The percentage of sulphur in an organic compound whose 0.32g produces 0.233g of  $BaSO_4$  [At.wt.Ba = 137, S=32] is  
(A) 1.0 (B) 10.0 (C) 23.5 (D) 32.1  
**CED366**
- The desiccants used for absorbing water during Liebig's method for estimation of carbon and hydrogen are  
(A) anhydrous  $CaCl_2$  (B) anhydrous  $Na_2SO_4$   
(C)  $Mg(ClO_4)_2$  (D)  $MgSO_4 \cdot 7H_2O$   
**CED367**

## Qualitative Analysis

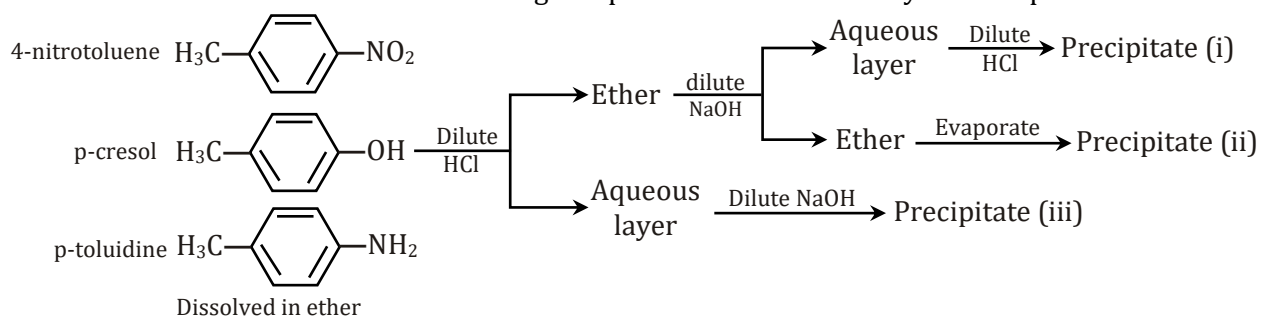
- The prussian blue colour obtained during the test of nitrogen by Lassaigne's test is due to the formation of :  
(A)  $Fe[Fe(CN)_6]_3$  (B)  $Na_3[Fe(CN)_6]$  (C)  $Fe(CN)_3$  (D)  $Fe_4[Fe(CN)_6]_3$   
**CED368**
- A compound which does not give a positive test in Lassaigne's test for nitrogen is:  
(A) urea (B) hydrazine (C) azobenzene (D) phenyl hydrazine  
**CED369**
- Disodium pentacyanonitrosyl ferrate reagent gives violet colour when which following element is present?  
(A) N (B) Cl (C) F (D) S  
**CED370**

## Separation Methods

- In the following extraction procedure, choose the number where nicotine would be found.



8. Choose the answer that has the following compounds located correctly in the separation scheme.



- (A) i=4-nitrotoluene, ii = p-cresol, iii = p-toluidine  
 (B) i=4-nitrotoluene, ii = p-toluidine, iii = p-cresol  
 (C) i=p-toluidine, ii = 4-nitrotoluene, iii = p-cresol  
 (D) i=p-cresol, ii = 4-nitrotoluene, iii = p-toluidine

CED372

ANSWER KEY

| Question | 1 | 2 | 3  | 4 | 5 | 6 | 7 | 8 |
|----------|---|---|----|---|---|---|---|---|
| Answer   | B | B | AC | D | B | D | C | D |